TRIGONOMETRIC SUBSTITUTION

The method of *trigonometric substitution* can be very effective in dealing with integrals when the integrands contain algebraic expressions such as $(a^2 - u^2)^{1/2}$, $(u^2 - a^2)^{3/2}$, and $1/(a^2 + u^2)^2$. There are three basic trigonometric substitutions:

Figure 7.5 and Table 7.4 summarize the three basic trigonometric substitutions for real numbers $a > 0$.

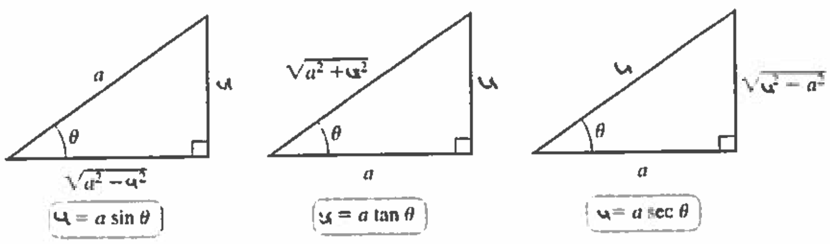

Figure 7.5

Table 7.4

The Integral Contains . . .	**Corresponding Substitution**	**Useful Identity**	
$a^2 - u^2$	$u = a\sin\theta,\ -\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$, for $\lvert u\rvert \le a$	$a^2 - a^2\sin^2\theta = a^2\cos^2\theta$	(i.e. $\sin^2\theta + \cos^2\theta = 1$)
$a^2 + u^2$	$u = a\tan\theta,\ -\frac{\pi}{2} < \theta < \frac{\pi}{2}$	$a^2 + a^2\tan^2\theta = a^2\sec^2\theta$	(i.e. $\tan^2\theta + 1 = \sec^2\theta$)
$u^2 - a^2$	$u = a\sec\theta,\ \begin{cases} 0 \le \theta < \frac{\pi}{2}, \text{ for } u \ge a \\ \frac{\pi}{2} < \theta \le \pi, \text{ for } u \le -a \end{cases}$	$a^2\sec^2\theta - a^2 = a^2\tan^2\theta$	(i.e. $\sec^2\theta - 1 = \tan^2\theta$)

What inspired these trig. change of variables?

The "sum of squares" forms in table 7.4 are difficult to integrate depending on their location and other ingrediants in the integrand. However, "sums of squares" can be re-written as a product of squares (which is easier to work with) .... think of the Pythagorean Thm: $a^2+b^2=c^2$. A rearrangement of this thm leads to the standard subst. for integrals involving the difference of squares a^2-u^2 (or u^2-a^2). The term $\sqrt{a^2-u^2}$ (respectively $\sqrt{u^2-a^2}$) is the length of one side of a right angled triangle whose hypotenuse has length a and whose other side has length u. Labeling one acute angle θ, we see that $u = a\sin\theta$. See diagram 1. Figure 7.5 above.

$\therefore\ a^2 - u^2 = a^2 - (a\sin\theta)^2 = a^2 - a^2\sin^2\theta = a^2(1-\sin^2\theta) = a^2\cos^2\theta$; "a^2-u^2" is now written as a product

Note :

- What we mean by the substitution $u = a\sin\theta$ is, more precisely, the *inverse* trigonometric substitution

$$\theta = \sin^{-1}\frac{u}{a}, \qquad -\frac{\pi}{2} \leqq \theta \leqq \frac{\pi}{2},$$

where $|u| \leqq a$. Suppose, for example, that an integral contains the expression $(a^2 - u^2)^{1/2}$. Then this substitution yields

$$(a^2 - u^2)^{1/2} = (a^2 - a^2\sin^2\theta)^{1/2} = (a^2\cos^2\theta)^{1/2} = a\cos\theta.$$

We chose the nonnegative square root in the last step because $\cos\theta \geqq 0$ for $-\pi/2 \leqq \theta \leqq \pi/2$. Thus the troublesome factor $(a^2 - u^2)^{1/2}$ becomes $a\cos\theta$ and, meanwhile, $du = a\cos\theta\, d\theta$.

- What we mean by the substitution $u = a\tan\theta$ in an integral that contains $a^2 + u^2$ is the substitution

$$\theta = \tan^{-1}\frac{u}{a}, \qquad -\frac{\pi}{2} < \theta < \frac{\pi}{2}.$$

In this case

$$\sqrt{a^2 + u^2} = \sqrt{a^2 + a^2\tan^2\theta} = \sqrt{a^2\sec^2\theta} = a\sec\theta,$$

under the assumption that $a > 0$. We take the positive square root in the last step here because $\sec\theta > 0$ for $-\pi/2 < \theta < \pi/2$.

- What we mean by the substitution $u = a\sec\theta$ in an integral that contains $u^2 - a^2$ is the substitution

$$\theta = \sec^{-1}\frac{u}{a}, \quad 0 \leqq \theta \leqq \pi,$$

where $|u| \geqq a > 0$ (because of the domain and range of the inverse secant function). Then

$$\sqrt{u^2 - a^2} = \sqrt{a^2\sec^2\theta - a^2} = \sqrt{a^2\tan^2\theta} = \pm a\tan\theta.$$

Here we must take the plus sign if $u > a$, so that $0 < \theta < \pi/2$ and $\tan\theta > 0$. If $u < -a$, so $\pi/2 < \theta < \pi$ and $\tan\theta < 0$, we take the minus sign.